

Comparative Review of Medical Imaging Datasets

This document presents a comparative overview of two widely used medical imaging datasets. Instead of listing information in the original format, the datasets are discussed from the perspective of research utility, data characteristics, and machine learning challenges.

Dataset A: NIH Chest X-ray14

The NIH Chest X-ray14 collection is one of the largest publicly available chest radiography resources. It contains 112,120 X-ray images representing a broad spectrum of thoracic conditions. The dataset includes categories such as Atelectasis, Cardiomegaly, Effusion, Infiltration, Mass, Nodule, Pneumonia, Pneumothorax, Edema, Fibrosis, Emphysema, Pleural Thickening, Hernia, and normal cases labeled as 'No Finding'.

Research Perspective: The dataset is valuable for developing computer-aided diagnostic systems because it offers a large number of patient scans and supports multi-label disease prediction.

Key Limitations: Disease categories are not uniformly distributed. Certain conditions are rare, resulting in limited training examples. Furthermore, some labels were generated automatically, which may introduce annotation inaccuracies. Another challenge is that a single image may contain multiple pathologies simultaneously.

Dataset B: HAM10000 Skin Lesion Collection

HAM10000 is a dermatology image dataset consisting of 10,015 dermoscopic images. It was designed to support the classification of skin lesions and includes diagnostic categories such as Melanoma, Melanocytic Nevi, Basal Cell Carcinoma, Benign Keratosis, Dermatofibroma, Vascular Lesions, and Actinic Keratoses.

Research Perspective: The dataset is frequently used for training deep learning models aimed at early skin cancer detection and automated dermatological screening.

Major Challenges: The image distribution is highly skewed toward the Melanocytic Nevi class. Researchers must also handle artifacts such as hair occlusions, lighting inconsistencies, and image quality variations that can affect model performance.

Overall Comparison

Although both datasets are widely adopted in medical AI research, they address different diagnostic domains. NIH Chest X-ray14 focuses on thoracic disease recognition from radiographic images, whereas HAM10000 concentrates on dermatological lesion analysis. Both datasets present class imbalance issues, making data preprocessing, augmentation, and balanced learning strategies important for achieving robust predictive performance.